

Uji Akurasi

Sample 1

Classification Metrics

...

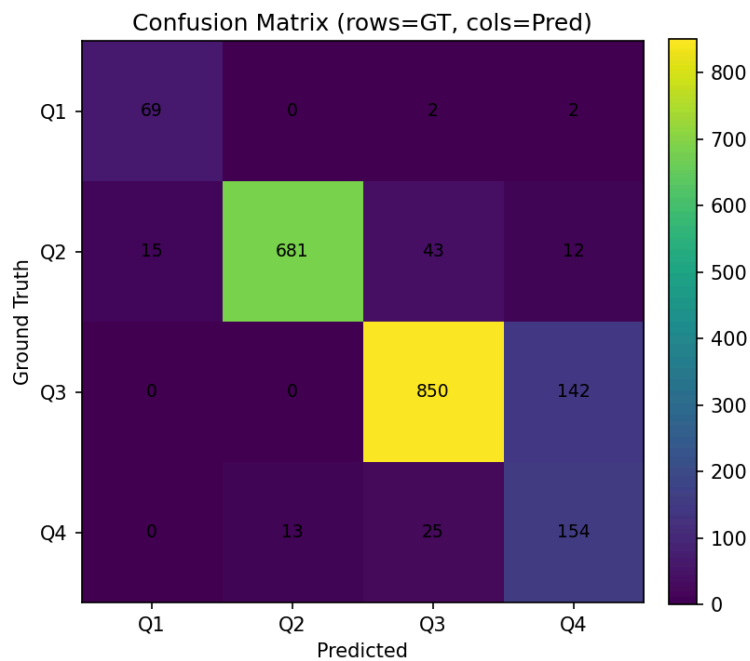
```
{
  "per_class": {
    "Q1": {
      "precision": 0.8214285714285714,
      "recall": 0.9452054794520548,
      "f1": 0.8789808917197452,
      "support": 73
    },
    "Q2": {
      "precision": 0.9812680115273775,
      "recall": 0.9067909454061251,
      "f1": 0.942560553633218,
      "support": 751
    },
    "Q3": {
      "precision": 0.9239130434782609,
      "recall": 0.8568548387096774,
      "f1": 0.8891213389121339,
      "support": 992
    },
    "Q4": {
      "precision": 0.4967741935483871,
      "recall": 0.8020833333333334,
      "f1": 0.6135458167330677,
      "support": 192
    }
  },
  "averages": {
    "micro": {
      "precision": 0.8735059760956175,
      "recall": 0.8735059760956175,
      "f1": 0.8735059760956175
    },
    "macro": {
      "precision": 0.8058459549956491,
      "recall": 0.8777336492252977,
      "f1": 0.8310521502495412
    },
    "weighted": {
```

```

    "precision": 0.9007962881788203,
    "recall": 0.8735059760956175,
    "f1": 0.8823893156811126
  }
},
"accuracy": 0.8735059760956175,
"support_total": 2008
}
...

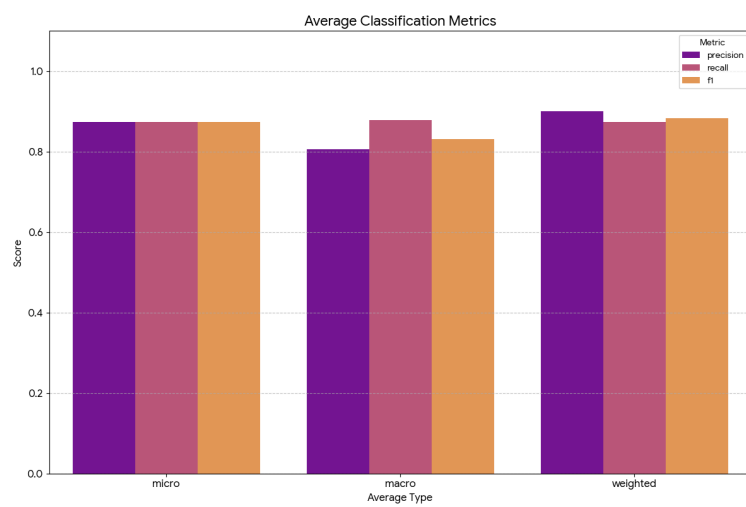
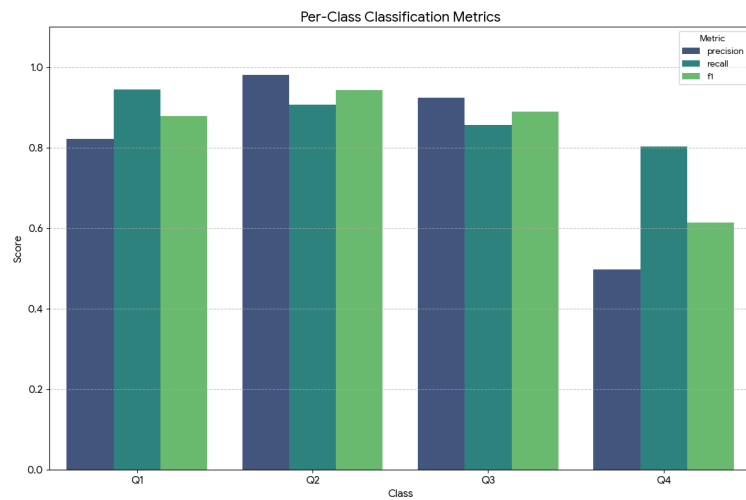
```

Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	69	0	2	2
Q2	15	681	43	12
Q3	0	0	850	142
Q4	0	13	25	154

Precision, Recall and F1



Sample 2

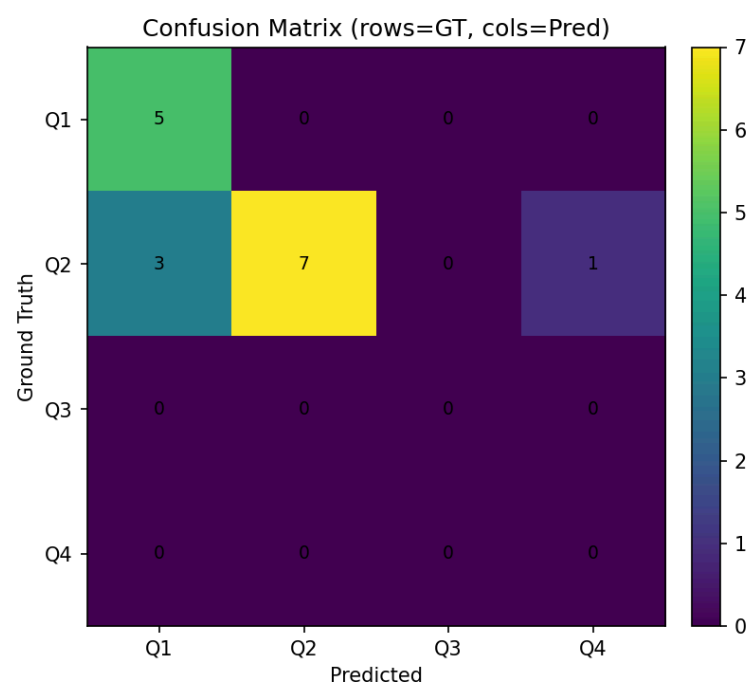
Classification Metrics

...

```
{
  "per_class": {
    "Q1": {
      "precision": 0.625,
      "recall": 1.0,
      "f1": 0.7692307692307693,
      "support": 5
    },
    "Q2": {
      "precision": 1.0,
      "recall": 0.6363636363636364,
      "f1": 0.7777777777777778,
      "support": 11
    },
    "Q3": {
```

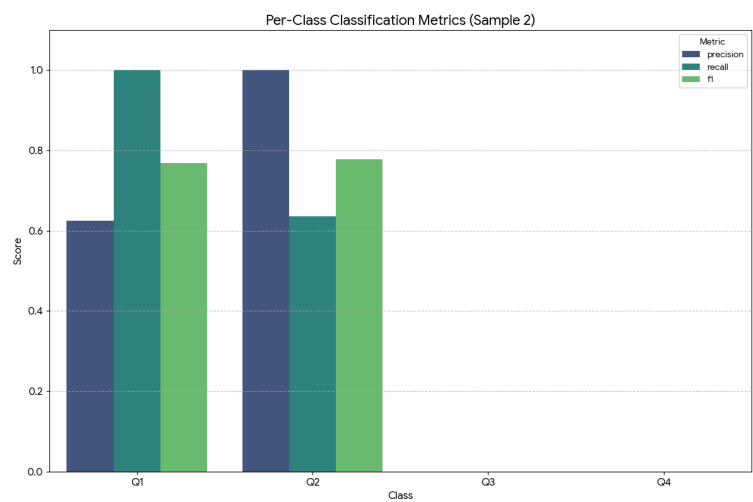
```
    "precision": 0.0,  
    "recall": 0.0,  
    "f1": 0.0,  
    "support": 0  
  },  
  "Q4": {  
    "precision": 0.0,  
    "recall": 0.0,  
    "f1": 0.0,  
    "support": 0  
  }  
},  
"averages": {  
  "micro": {  
    "precision": 0.75,  
    "recall": 0.75,  
    "f1": 0.75  
  },  
  "macro": {  
    "precision": 0.40625,  
    "recall": 0.40909090909090906,  
    "f1": 0.38675213675213677  
  },  
  "weighted": {  
    "precision": 0.8828125,  
    "recall": 0.75,  
    "f1": 0.7751068376068376  
  }  
},  
"accuracy": 0.75,  
"support_total": 16  
}  
...
```

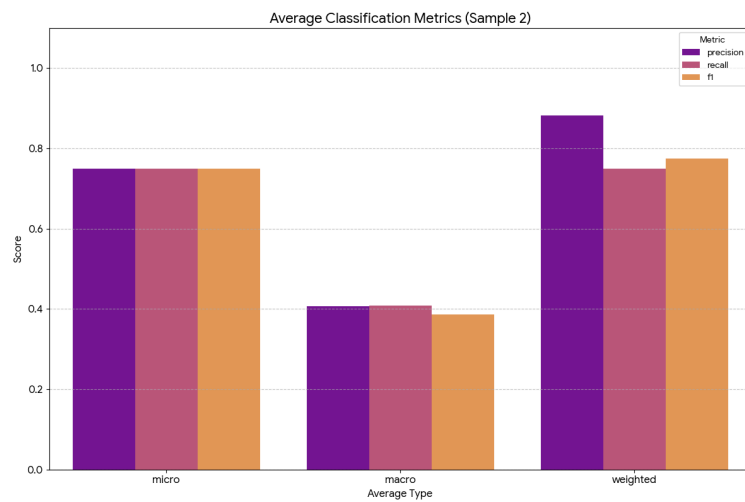
Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	5	0	0	0
Q2	3	7	0	1
Q3	0	0	0	0
Q4	0	0	0	0

Precision, Recall and F1





Sample 3

Classification Metrics

...

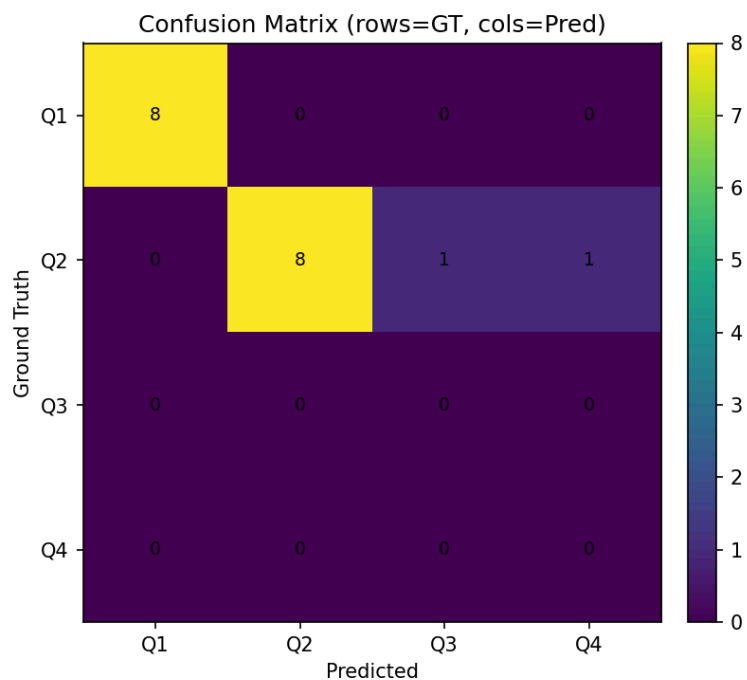
```
{
  "per_class": {
    "Q1": {
      "precision": 1.0,
      "recall": 1.0,
      "f1": 1.0,
      "support": 8
    },
    "Q2": {
      "precision": 1.0,
      "recall": 0.8,
      "f1": 0.8888888888888889,
      "support": 10
    },
    "Q3": {
      "precision": 0.0,
      "recall": 0.0,
      "f1": 0.0,
      "support": 0
    },
    "Q4": {
      "precision": 0.0,
      "recall": 0.0,
      "f1": 0.0,
      "support": 0
    }
  },
  "averages": {
```

```

"micro": {
  "precision": 0.8888888888888888,
  "recall": 0.8888888888888888,
  "f1": 0.8888888888888888
},
"macro": {
  "precision": 0.5,
  "recall": 0.45,
  "f1": 0.4722222222222222
},
"weighted": {
  "precision": 1.0,
  "recall": 0.8888888888888888,
  "f1": 0.9382716049382717
}
},
"accuracy": 0.8888888888888888,
"support_total": 18
}
...

```

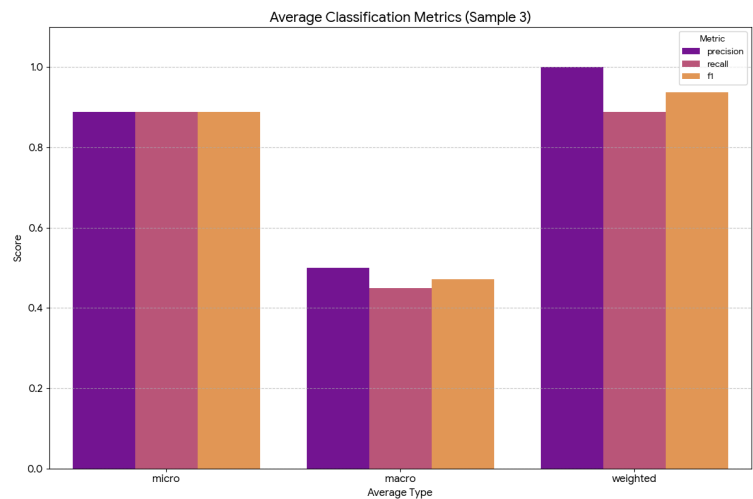
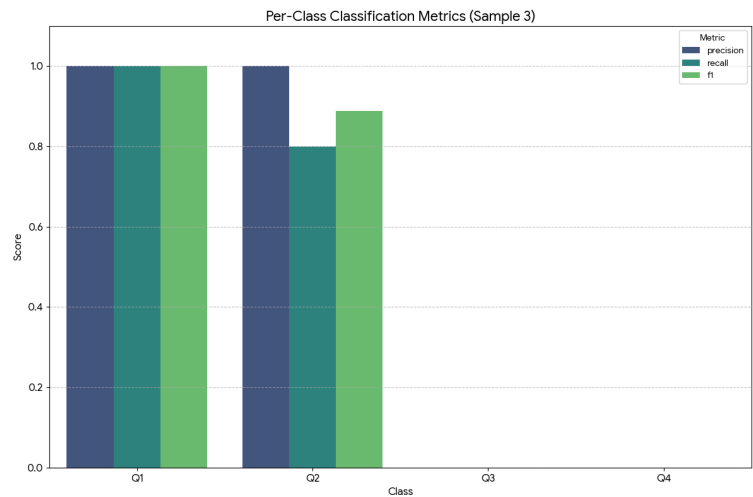
Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	8	0	0	0
Q2	0	8	1	1

Q3	0	0	0	0
Q4	0	0	0	0

Precision, Recall and F1



Sample 5

Classification Metrics

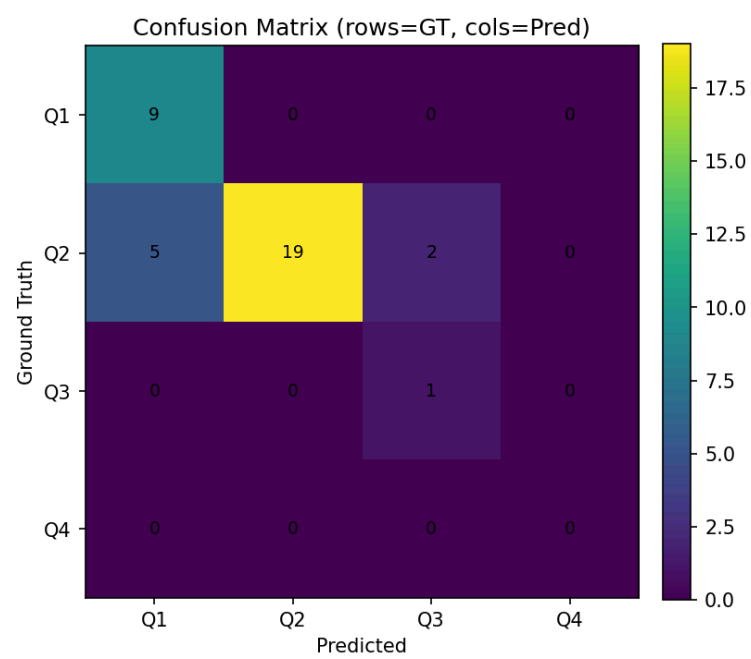
...

```
{
  "per_class": {
    "Q1": {
      "precision": 0.6428571428571429,
      "recall": 1.0,
      "f1": 0.782608695652174,
      "support": 9
    }
  }
}
```



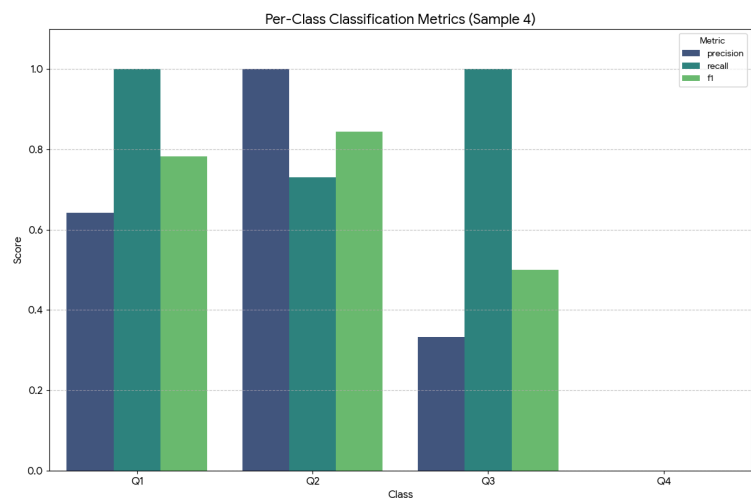
```
},
"Q2": {
  "precision": 1.0,
  "recall": 0.7307692307692307,
  "f1": 0.8444444444444443,
  "support": 26
},
"Q3": {
  "precision": 0.3333333333333333,
  "recall": 1.0,
  "f1": 0.5,
  "support": 1
},
"Q4": {
  "precision": 0.0,
  "recall": 0.0,
  "f1": 0.0,
  "support": 0
}
},
"averages": {
  "micro": {
    "precision": 0.8055555555555556,
    "recall": 0.8055555555555556,
    "f1": 0.8055555555555556
  },
  "macro": {
    "precision": 0.494047619047619,
    "recall": 0.6826923076923077,
    "f1": 0.5317632850241546
  },
  "weighted": {
    "precision": 0.8921957671957672,
    "recall": 0.8055555555555556,
    "f1": 0.8194176060118088
  }
},
"accuracy": 0.8055555555555556,
"support_total": 36
}
...
```

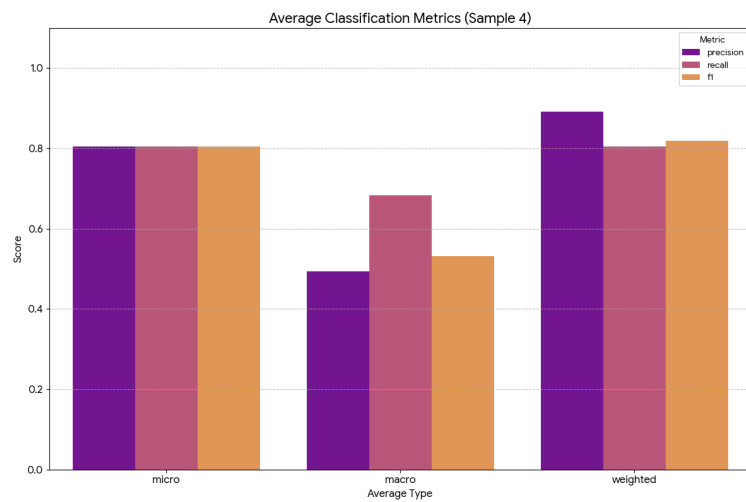
Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	9	0	0	0
Q2	5	19	2	0
Q3	0	0	1	0
Q4	0	0	0	0

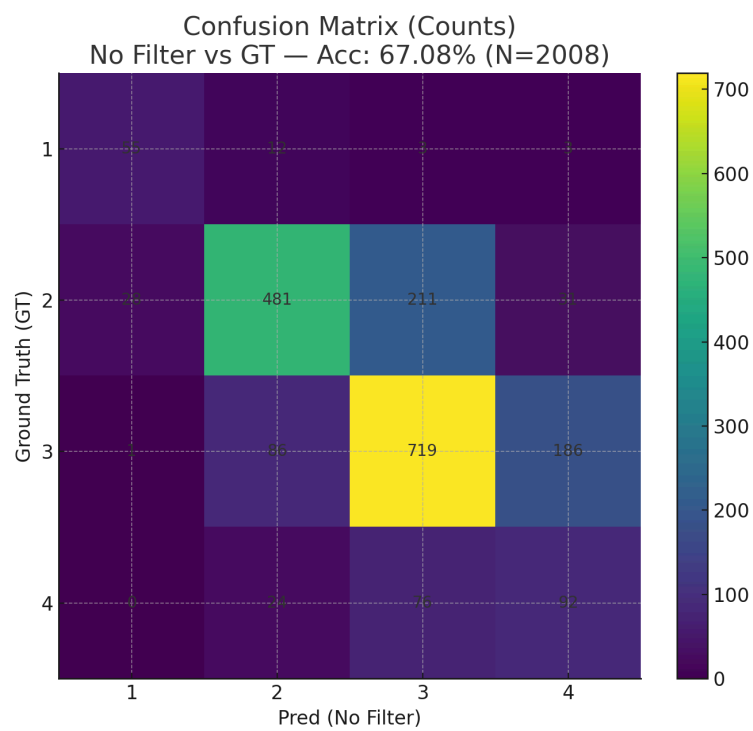
Precision, Recall and F1





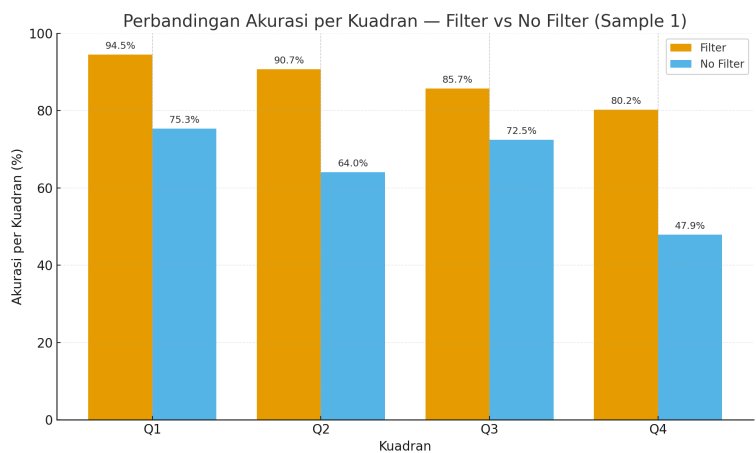
Sample 01 No Filter

Confusion Matrix



GT/Pred	1	2	3	4
1	55	12	3	3
2	28	481	211	31
3	1	86	719	186
4	0	24	76	92

Comparison



Quadra nt	Recall_Filter_ %	Recall_NoFilter_ %	Diff_NoFilter_minus_Filter_ pp	GT_Support_ N
Q1	94.52	75.34	-19.18	73
Q2	90.68	64.05	-26.63	751
Q3	85.69	72.48	-13.21	992
Q4	80.21	47.92	-32.29	192

Uji Ketahanan dan Keandalan

CPU

CPU 1 Camera Pstats

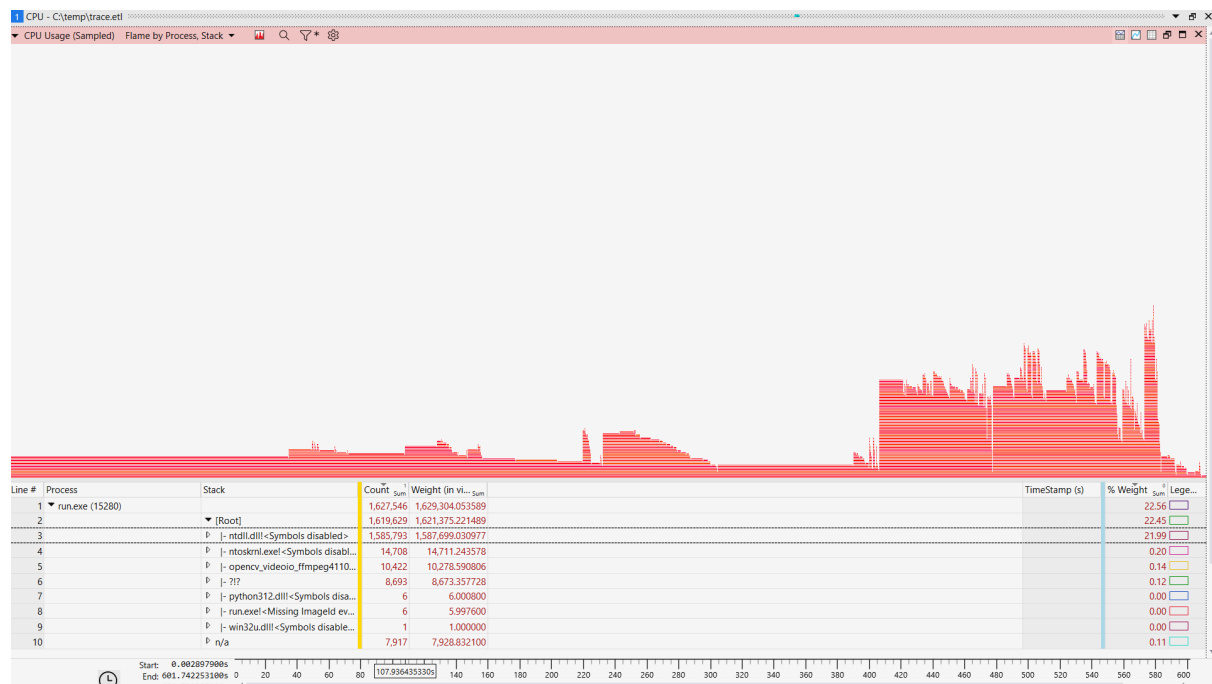
func_label	ncall s	cumtim e (s)	percall (ms)	grou p	description
stdlib/queue.py - get	71	73.5	13.96	stdli b	Ambil item dari Queue secara blocking
stdlib/~ - <built-in method builtins.exec>	191	56	0.01	stdli b	Interpreter mengeksekusi string kode Python.

stdlib/___init___.py - <module>	1	56	16.71	stdli b	Inisialisasi paket utama aplikasi EyeLog.
stdlib/~ - <built-in method exec>	1	54.4	9577.5 3	stdli b	Fungsi bawaan untuk menjalankan blok kode dinamis.
stdlib/threading.py - wait	81	45.3	3.35	stdli b	Thread menunggu Event/Condition hingga dipicu atau timeout.
app/camera_detail.py - closeEvent	1	35.5	0.08	app	Cleanup lengkap saat jendela detail kamera ditutup, termasuk stop StreamWorker.
app/coverage_logger.py - stop	1	35.5	1.59	app	Menghentikan logger asinkron dan flush CSV log coverage.
stdlib/threading.py - join	1	35.5	0.04	stdli b	Menunggu thread target selesai sebelum melanjutkan.
stdlib/~ - <method 'join' of '_thread._ThreadHandle' objects>	1	35.5	0.24	stdli b	Lower level join handle thread di C API.
stdlib/threading.py - _bootstrap	1	35.5	0.01	stdli b	Rutinitas internal untuk memulai thread Python.
stdlib/threading.py - _bootstrap_inner	1	35.5	0.07	stdli b	Bagian inti bootstrap yang menjalankan target function.
app/stream_worker.py - _iteration	382	29.7	0.93	app	Loop periodik QTimer: baca & proses frame, cek state, handle reconnect.
dep/~ - <method 'acquire' of '_thread.lock' objects>	360	28.1	23.51	dep	Mengunci mutex/kunci thread safe.
dep/~ - <built-in method msleep>	115	9.5	61.09	dep	Tidur dalam milidetik di thread Qt (QThread.msleep).
app/stream_worker.py - _process_frame	250	8.9	1.68	app	Pipeline frame + ekstraksi foreground +

					kontur + metrik coverage.
app/roi_window.py - update_preview	1406	8.6	0.13	app	Refresh tampilan pratinjau ROI di overlay UI.
dep/~ - <method 'exec' of 'PySide6.QtWidgets.QDialog' objects>	4	4.7	1156.69	dep	Menampilkan dialog Qt secara modal & tunggu respon.
app/material_detector.py - process_frame	250	4.1	0.32	app	Lakukan backgroundâ subtraction + morfologi untuk hasilkan mask biner.
app/stream_worker.py - _read_with_timeout	266	3.8	0.03	app	Membaca frame cv2.VideoCapture dengan batas waktu aman.
dep/~ - <built-in method information>	6	3.6	512.71	dep	QMessageBox.information kotak dialog info.
app/main_window.py - _on_edit_camera_data_loaded	1	3.5	0.09	app	Callback: buka dialog edit setelah data kamera di fetch
dep/~ - <built-in method fromImage>	1656	3.1	1.89	dep	Konversi QImage QPixmap (Qt).
dep/~ - <method 'copy' of 'numpy.ndarray' objects>	2760	3	0.96	dep	Duplikasi matriks NumPy tanpa pengaruh aslinya.
dep/~ - <cvtColor>	2406	2.5	0.74	dep	OpenCV cvtColor konversi ruang warna.
dep/~ - <method 'apply' of 'cv2.BackgroundSubtractorMOG2' objects>	250	2.4	7.02	dep	OpenCV MOG2 apply() hasilkan mask foreground.
app/material_detector.py - process_mask	250	2.3	0.2	app	Analisis mask biner hitung kontur & statistik coverage.
app/preview_scheduler.py - run	1	2	0.14	app	Thread SnapshotWorker:

					jadwalkan & ambil snapshot kamera.
app/preview_scheduler.py - get_capture	1	2	0.03	app	Ambil VideoCapture dari pool reuse atau buat baru.
app/preview_scheduler.py - _create_capture	1	2	2003.2	app	Buat VideoCapture baru dengan pengaturan spesifik.
app/roi_selector.py - _update_display	1416	1.9	0.04	app	Render ulang ROI yang dipilih ke layar selama seleksi.

CPU 3 Camera Trace Windows Performance Record



Process and Stack

Line #	Process	Stack	Count	Weight (in view) (ms)	Timestamp (s)	% Weight
1	run.exe (15280)		1690396	1,692,241.046189		22.04

2		[Root]	16798 85	1,681,715.057 789		21.90
3		- ntdll.dll!<Symbols disabled>	16453 64	1,647,355.232 077		21.45
4		- ntoskrnl.exe!<Symbols disabled>	15181	15,185.16297 8		0.20
5		- opencv_videoio_ffmpeg4110_64.dll!<Missing ImageIeld event>	10532	10,386.99480 6		0.14
6		- ?!?	8795	8,774.669528		0.11
7		- python312.dll!<Symbols disabled>	6	6.000800		0.00
8		- run.exe!<Missing ImageIeld event>	6	5.997600		0.00
9		- win32u.dll!<Symbols disabled>	1	1.000000		0.00
10		n/a	10511	10,525.98840 0		0.14

Process and Thread



Line #	Process	Thread ID	Count	Weight (in view) (ms)	TimeStamp (s)	% Weight
1	run.exe (15280)		1690396	1,692,241.046189		22.04
2		10460	244187	244,007.691436		3.18
3		10320	68837	68,973.300823		0.90
4		7508	80176	80,327.930273		1.05
5		13060	76152	76,313.531064		0.99
6		10484	38891	38,975.321707		0.51
7		12204	77714	77,869.376764		1.01
8		8864	1205	1,193.365896		0.02
9		20172	79661	79,803.584648		1.04
10		9956	39368	39,464.853514		0.51
11		14720	79072	79,254.102036		1.03
12		17412	79840	79,995.884066		1.04
13		7296	74667	74,811.203328		0.97
14		21156	80569	80,716.585089		1.05
15		14188	39375	39,450.110230		0.51
16		9924	77692	77,837.016924		1.01
17		12684	78099	78,252.629507		1.02
18		10008	39214	39,289.390097		0.51
19		19788	39111	39,183.180095		0.51
20		15204	76542	76,687.844463		1.00
21		3804	80481	80,638.787711		1.05
22		12540	39362	39,435.875880		0.51
23		14572	29	28.963000		0.00
24		13092	116652	116,455.810316		1.52

25		11624	1292	1,275.867597		0.02
26		5448	10	10.000000		0.00
27		9932	23	22.964900		0.00
28		4724	18	17.953000		0.00
29		10604	202	199.701800		0.00
30		17500	27	26.985200		0.00
31		16944	18	17.745700		0.00
32		7344	1228	1,210.610394		0.02
33		11228	1321	1,305.205205		0.02
34		15812	1061	1,063.669493		0.01
35		16556	18	18.156300		0.00
36		2264	12	12.015900		0.00
37		10628	4	4.000000		0.00
38		20256	3	3.000000		0.00
39		18800	13	13.048200		0.00
40		18364	126	125.871500		0.00
41		15608	5	5.001200		0.00
42		17652	18	18.027200		0.00
43		12340	1276	1,258.607015		0.02
44		8768	1	1.000000	332.3552975 00	0.00
45		18168	4	3.982200		0.00
46		14128	178	177.615800		0.00
47		9724	970	971.713800		0.01
48		5472	4	4.015200		0.00
49		5932	3	2.971800		0.00
50		15144	198	197.699100		0.00

51		16292	1433	1,416.780294		0.02
52		15520	20	19.850800		0.00
53		4020	190	189.717600		0.00
54		8436	218	217.771200		0.00
55		11568	18	17.937900		0.00
56		21216	3	3.000000		0.00
57		12568	1424	1,409.363404		0.02
58		16784	24	23.960700		0.00
59		20024	1272	1,256.442305		0.02
60		14532	1283	1,267.473603		0.02
61		20196	1	1.061200	618.2089573 00	0.00
62		20688	1919	1,920.048000		0.03
63		6640	11	11.018800		0.00
64		14924	9	8.979500		0.00
65		6872	13	13.028100		0.00
66		13848	6	6.012200		0.00
67		4372	358	358.181700		0.00
68		16976	3001	3,003.423600		0.04
69		7000	5	5.016000		0.00
70		20268	1266	1,248.264310		0.02
71		18988	13	12.995900		0.00
72		5040	7	7.000000		0.00
73		11220	1272	1,263.539106		0.02
74		17332	66	65.121500		0.00
75		3792	5	5.000000		0.00
76		10308	5	5.000000		0.00

77		14560	4	4.010500		0.00
78		8496	18	18.027600		0.00
79		8396	5	5.008800		0.00
80		12644	5	5.002400		0.00
81		6180	15	14.987400		0.00
82		12696	3	3.065500		0.00
83		5752	4	4.009200		0.00
84		13244	190	189.513800		0.00
85		12980	22	22.162300		0.00
86		21372	187	184.579800		0.00
87		6484	11	11.000000		0.00
88		21192	38	37.941200		0.00
89		10144	9	8.939100		0.00
90		12488	1	1.000000	332.3618682 00	0.00
91		10476	1282	1,267.694597		0.02
92		17900	8	8.010000		0.00
93		20552	10	10.009500		0.00
94		21308	7	7.011200		0.00
95		10304	178	177.737100		0.00
96		20336	22	21.894000		0.00
97		17624	23	22.960800		0.00
98		7776	4	4.000000		0.00
99		20528	2980	2,980.420000		0.04
100		20488	13	13.000000		0.00
101		15480	15644	15,652.659800		0.20
102		11128	24	23.957100		0.00

103		17852	20	20.037100		0.00
104		7632	9	9.000000		0.00
105		9132	77	77.119500		0.00
106		7044	17467	17,455.359600		0.23
107		16988	22	22.081400		0.00
108		3440	167	166.399100		0.00
109		2072	168	166.895500		0.00
110		17144	20	19.993600		0.00
111		16932	20	19.947100		0.00
112		7496	18	18.008500		0.00
113		12572	201	198.830300		0.00
114		21168	20	19.881300		0.00
115		15140	17	16.933000		0.00
116		8212	22	21.986100		0.00
117		14400	20	19.974300		0.00
118		12956	22	21.989500		0.00
119		15688	2	2.000000		0.00
120		15592	24	24.027900		0.00
121		13552	25	24.971600		0.00
122		15220	114	113.904699		0.00
123		16176	19	19.002100		0.00
124		20812	27	27.064000		0.00
125		17256	20	19.964100		0.00
126		7912	9	9.098400		0.00
127		5456	9	9.002800		0.00
128		17040	2	2.034500		0.00
129		8068	3	2.995400		0.00

130		10224	19	18.969800		0.00
131		20948	4	4.000000		0.00
132		1656	21	20.975800		0.00
133		15540	18	17.972300		0.00
134		16656	162	161.685800		0.00
135		2168	8	7.852800		0.00
136		10300	17	16.987600		0.00
137		17172	20	20.010600		0.00
138		6552	215	214.061300		0.00
139		2468	229	229.007100		0.00
140		11500	6	6.006200		0.00
141		19376	179	178.849300		0.00
142		20604	16	15.940400		0.00
143		12532	184	183.687700		0.00
144		5068	212	211.772800		0.00
145		11092	32	32.193800		0.00
146		14432	197	194.371800		0.00
147		7468	1	1.000000	378.2869066 00	0.00
148		5476	7	7.128900		0.00
149		1232	15	14.994100		0.00
150		3464	4	4.000000		0.00
151		15988	198	196.299800		0.00
152		14004	16	15.880600		0.00
153		7436	12	12.002900		0.00
154		9728	30	30.045500		0.00
155		872	19	18.754000		0.00

156		17924	11	10.911400		0.00
157		14536	6	6.004800		0.00
158		7052	21	21.001000		0.00
159		17316	7	7.067500		0.00
160		1848	5	5.007900		0.00
161		19316	3	3.000000		0.00
162		12648	2	2.000000		0.00
163		18024	3	3.000000		0.00
164		9588	16	15.997400		0.00
165		12820	26	26.034400		0.00
166		11136	8	7.996500		0.00
167		3316	20	19.937700		0.00
168		2580	18	17.955200		0.00
169		17204	178	177.613900		0.00
170		1680	144	143.206400		0.00
171		18460	184	181.041900		0.00
172		9476	16753	16,737.115500		0.22
173		6288	22	21.991800		0.00
174		18716	208	206.367600		0.00
175		2672	154	152.531400		0.00
176		19920	138	137.133300		0.00
177		19252	16	16.048400		0.00
178		19244	25	24.991100		0.00
179		16840	14	14.124100		0.00
180		20940	173	172.087800		0.00
181		20440	22	21.835800		0.00
182		4180	150	148.656200		0.00

183		2396	20	20.042400		0.00
184		15008	164	162.669000		0.00
185		9860	25	25.005300		0.00
186		21424	18	17.962700		0.00
187		7140	38	38.099600		0.00
188		20692	243	241.024400		0.00
189		14100	10	10.000000		0.00
190		10436	3	3.000000		0.00
191		2972	4	4.059400		0.00
192		20304	4	3.965700		0.00
193		14680	2	2.000000		0.00
194		4804	5	5.064600		0.00
195		6700	5	4.954200		0.00
196		11348	7	7.125300		0.00
197		20732	4	3.995500		0.00
198		1372	20	19.817000		0.00
199		8556	159	157.811700		0.00
200		15156	19	18.941500		0.00
201		11380	3	3.004800		0.00
202		12740	4	3.993800		0.00
203		12396	5	5.013700		0.00
204		2208	6	6.074800		0.00
205		19120	5	4.967600		0.00
206		14616	3	3.034600		0.00
207		18160	4	3.972000		0.00
208		4044	180	178.041100		0.00
209		5212	2	2.000000		0.00

210		19816	13	13.027500		0.00
211		19768	18	17.898300		0.00
212		10396	15	14.902200		0.00
213		5496	21	20.809300		0.00
214		21060	197	194.286800		0.00
215		17436	202	200.011500		0.00
216		10048	18	18.113500		0.00
217		2268	205	203.060000		0.00
218		20656	201	199.041200		0.00
219		6272	25	25.057900		0.00
220		1748	16	15.931200		0.00
221		1312	242	240.442700		0.00
222		15228	6	6.000000		0.00
223		9384	22	22.089300		0.00
224		19576	2	2.000000		0.00
225		11100	8	8.055900		0.00
226		15856	4	4.004700		0.00
227		17644	3	3.085100		0.00
228		432	6	5.775000		0.00
229		17472	3	3.008000		0.00
230		14816	5	5.000900		0.00
231		9792	11	10.836400		0.00
232		19948	1	1.000000	141.2290068 00	0.00

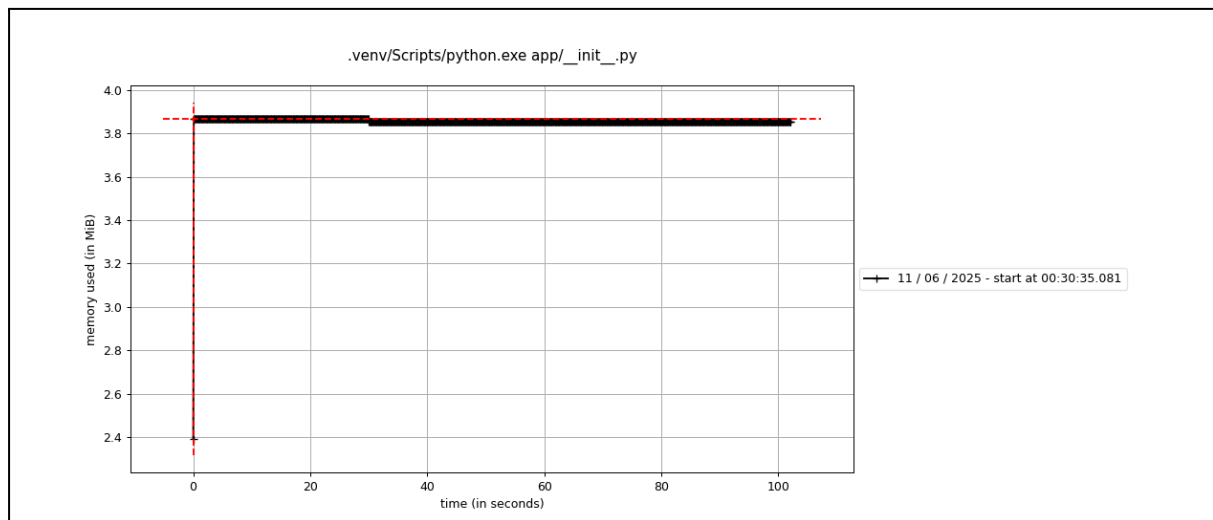
FPS

source	frames	total_time_s	fps	ms_per_frame
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./resource/video_dir/sample_01/ROI_01_CUT.mp4	37481	1249.367	30	33.33
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Memory

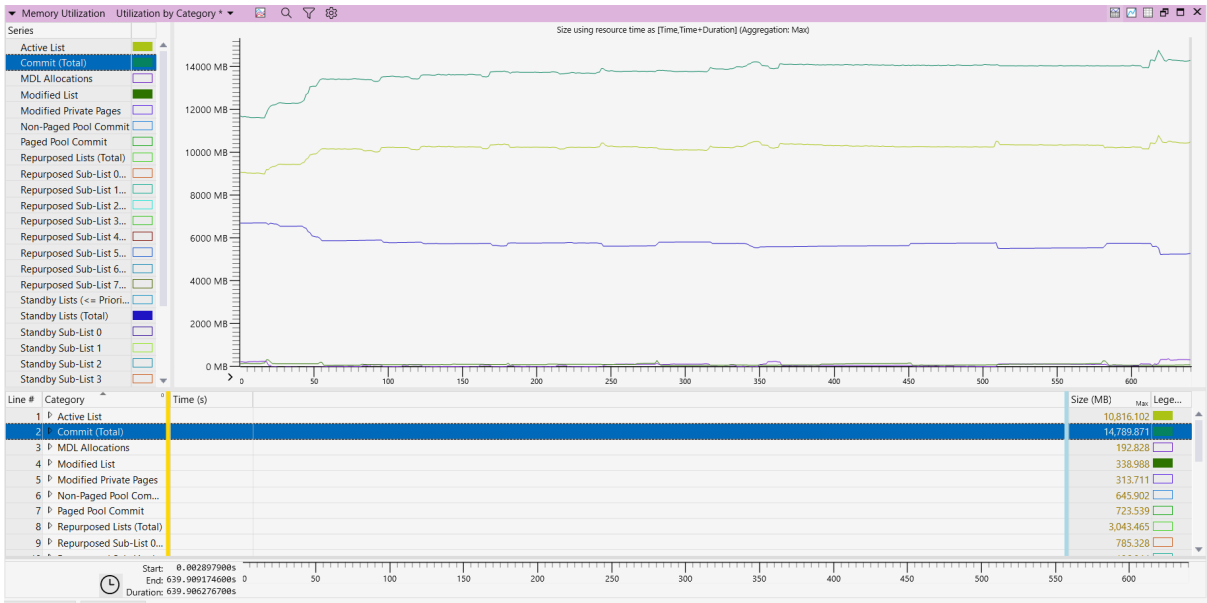
Memory 1 Camera Mprof



Profiling memori menggunakan mprof (interval sampling 0,1 detik) menunjukkan bahwa penggunaan memori aplikasi sangat ringan dan stabil. Selama periode pengujian 100 detik—yang mencakup proses startup, streaming video, pengolahan algoritma background subtraction MOG2, interaksi GUI, hingga proses penutupan—penggunaan RAM tercatat hanya sekitar **3,83 ± 0,02 MB**. Grafik moving-average penggunaan memori memperlihatkan kurva datar, menandakan tidak adanya tren kenaikan memori secara kumulatif yang mengindikasikan memory leak. Penggunaan memori tertinggi selama pengujian tetap di bawah 4 MB, jauh dari ambang batas ideal sekitar 100–200 MB untuk aplikasi desktop ringan. Setelah aplikasi ditutup, penggunaan memori kembali ke baseline interpreter Python (~2,3 MB)—menandakan bahwa semua alokasi memori seperti buffer video telah dilepas dengan sempurna.

Dengan profil memori seoptimal ini, aplikasi dapat dijalankan dengan aman bahkan pada perangkat keras lama seperti laptop dengan Intel Core i5-4300U dan RAM 8 GB, tanpa risiko masalah thrashing atau penggunaan swap berlebihan. Grafik dan data lengkap pengukuran memori dapat dilihat pada [memory_profile.png](#) dan [memory_profile.csv](#).

Memory 3 Camera Windows Performance Recorder



Line #	Category	Size (MB)
1	Active List	10,816.102
2	Commit (Total)	14,789.871
3	MDL Allocations	192.828
4	Modified List	338.988
5	Modified Private Pages	313.711
6	Non-Paged Pool Commit	645.902
7	Paged Pool Commit	723.539
8	Repurposed Lists (Total)	3,043.465
9	Repurposed Sub-List 0 (Delta)	785.328
10	Repurposed Sub-List 1 (Delta)	196.844
11	Repurposed Sub-List 2 (Delta)	354.766
12	Repurposed Sub-List 3 (Delta)	0.000
13	Repurposed Sub-List 4 (Delta)	145.926
14	Repurposed Sub-List 5 (Delta)	1,560.602
15	Repurposed Sub-List 6 (Delta)	0.000

16	Repurposed Sub-List 7 (Delta)	0.000
17	Standby Lists (<= Priority 5)	6,653.402
18	Standby Lists (Total)	6,731.398
19	Standby Sub-List 0	195.395
20	Standby Sub-List 1	34.500
21	Standby Sub-List 2	95.453
22	Standby Sub-List 3	0.000
23	Standby Sub-List 4	42.055
24	Standby Sub-List 5	6,617.613
25	Standby Sub-List 6	8.609
26	Standby Sub-List 7	78.223
27	Total Physical Memory	16,206.000
28	Zero and Free Lists	398.063

Unit Test

```
PC-Windows@DESKTOP-U88U1AD MINGW64 /c/Users/PC-Windows/Documents
/Github/opencv-py-pdu/app (master)
$ pytest
===== test session starts =====
platform win32 -- Python 3.13.2, pytest-8.4.0, pluggy-1.6.0
rootdir: C:\Users\PC-Windows\Documents\Github\opencv-py-pdu\app
collected 13 items

tests\test_camera_roi.py ..                [ 15%]
tests\test_camera_url.py ...               [ 38%]
tests\test_database.py .....              [ 76%]
tests\test_material_detector.py ...        [100%]

===== 13 passed in 0.58s =====
(.venv)
PC-Windows@DESKTOP-U88U1AD MINGW64 /c/Users/PC-Windows/Documents
/Github/opencv-py-pdu/app (master)
$ █
```

No	Nama Test
1	test_camera_roi_saves_correct_roi
2	test_camera_roi_remove
3	test_camera_url_validation_valid
4	test_camera_url_parse_host_and_port
5	test_insert_and_fetch_camera
6	test_foreground_detects_object
7	test_roi_handling_per_camera
8	test_contour_detection_on_foreground
9	test_foreground_blank_returns_zero
10	test_update_camera_data

Hardware Specifications

Device name	DESKTOP-U88U1AD
Processor	Intel(R) Core(TM) i5-10500H CPU @ 2.50GHz (2.50 GHz)
Installed RAM	16.0 GB (15.8 GB usable)
Device ID	9210D779-C194-41A5-B532-2F9172A60662
Product ID	00327-36350-74292-AAOEM
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display